

In the middle of the bridge

W24 (2024) — English translation

In this problem, we study electrical circuits containing a nonlinear element, the current-voltage characteristic of which is shown in Fig. 1. The current-voltage characteristic of a nonlinear element is described by two sections: $I = I_0 \cdot (U/U_0)$ at $U \leq U_0$, where $I_0 = 10$ mA and $U_0 = 5$ V. $I = I_0 = \text{const}$ at $U \geq U_0$.

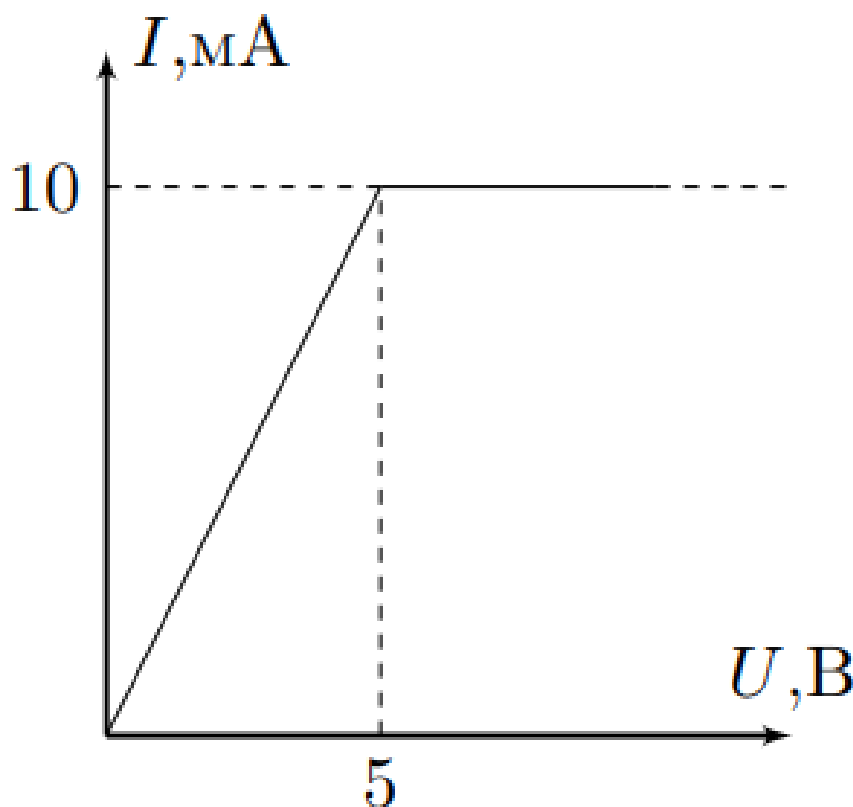


Figure 1: Figure 1.

Part A. DC Circuit (3.0 points)

Figure 2 shows a diagram of an electrical circuit consisting of resistors with resistances R , $2R$, $4R$, $6R$ and $8R$, an ideal constant voltage source $\mathcal{E}$ and the described nonlinear element. This part of the problem is devoted to studying the dependence of the current through a nonlinear element depending on the source voltage.

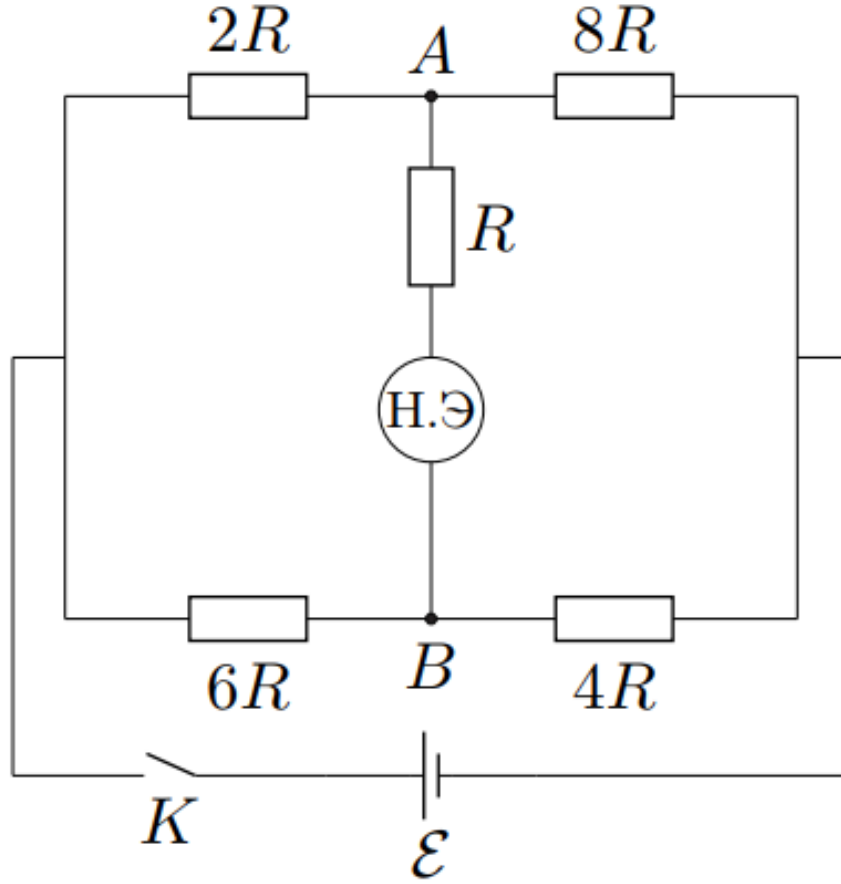


Figure 2: Figure 2.

- A1** Since without a section of the circuit between nodes A and B the electrical circuit consists entirely of constant voltage sources and resistors, the dependence of the current I_{AB} flowing from node A to node B on the voltage $U_{AB} = \varphi_A - \varphi_B$ in this section of the circuit can be represented in a form that does not depend on the element connected between nodes A and B : **1.50**

$$I_{AB} = \frac{\mathcal{E}_0 - U_{AB}}{r_0}.$$

Determine $\mathcal{E}_0$ and r_0 . Express your answers in terms of $\mathcal{E}$ and R .

- A2** Obtain the dependence of the current strength I_{AB} on the value $\mathcal{E}$. Express your answer using $\mathcal{E}$, R , U_0 and I_0 . **0.50**

- A3** Obtain the dependence of the power $P_{n,e}$ released on the nonlinear element on the value $\mathcal{E}$. Express your answer using $\mathcal{E}$, R , U_0 and I_0 . **0.50**

- A4** Construct a high-quality graph of the dependence $P_{n,e}(\mathcal{E})$. Indicate all the characteristic points on the graph and indicate their corresponding values. **0.50**

Part B. Electric Circuit Containing a Capacitor (5.0 points)

The figure shows an electrical circuit consisting of an ideal voltage source $\mathcal{E}$, a resistor $R = 500 \, \Omega$, a capacitor with a capacitance $C = 100 \, \mu\text{F}$ and a nonlinear element connected in series, as well as a switch K . Initially, the key is open and the capacitor is not charged. At the initial time $t = 0$ the key is closed.

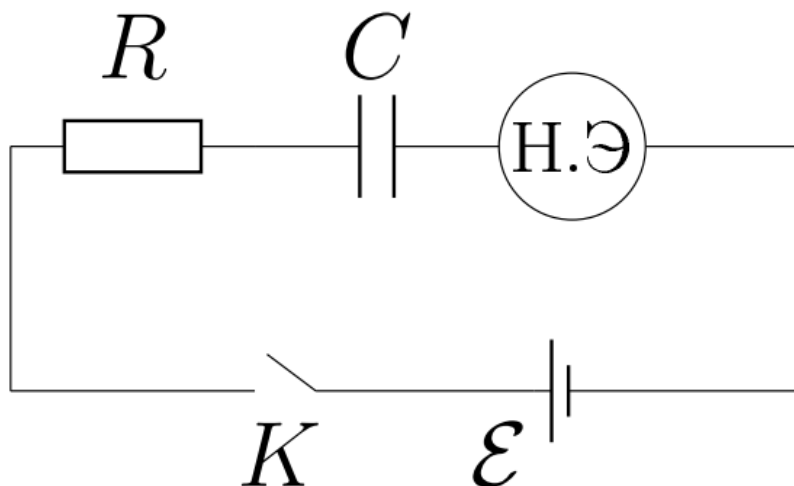


Figure 3: Figure 3.

Let the emf of the source be equal to $\mathcal{E} = 6 \, \text{V}$.

B1	Obtain the dependence of the current $I_C(t)$ through the capacitor on time t . Express your answer in terms of $\mathcal{E}$, R , U_0 , I_0 , C and t .	0.80
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B2	What amount of heat $Q_{\text{n.e}}$ will be released on the nonlinear element during the establishment of constant current? Express your answer in terms of $\mathcal{E}$, R , U_0 , I_0 and C . Find the numerical value.	0.60
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B3	After what time τ the amount of heat released on the nonlinear element will reach 80% from the value $Q_{\text{n.e}}$? Express your answer in terms of R , U_0 , I_0 , and C . Find the numerical value.	0.60
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Now the emf of the source is $\mathcal{E} = 12 \, \text{V}$.

B4	Obtain the dependence of the current $I_C(t)$ through the capacitor on time t . Express your answer in terms of $\mathcal{E}$, R , U_0 , I_0 , C and t .	1.50
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B5	What amount of heat $Q_{\text{n.e}}$ will be released on the nonlinear element during the establishment of constant current? Find the numerical value with an accuracy of at least 1%.	1.50
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Source: <https://pho.rs/p/3553>